Progress Report 3

Week 3: Baselines B1, B2 & B3, Evaluation Engine, and Beam Search

Cole Rodney Migisha	b29657	Charis Opol	b29654
Ainomugisha Faith	b27518	Lisa Lusinga Namugenyi	b30300
Ansasire Nehum Naamaye	b29785	Wakisa Isabelle	b30311

Project: LipScribe – Lip-Assisted Speech Recognition for Robust Live Captioning 5 November 2026

1 Summary

In Week 3 (30 October–5 November 2026), the team successfully completed all core deliverables for Milestones **M3** and **M4**. All three foundational baseline models—audio-only baseline **B1**, visual-only baseline **B2**, and concatenation fusion baseline **B3**—have completed full-corpus training on our dedicated Alienware GPU workstation and have been officially evaluated on the four held-out test speakers across all five standardised acoustic conditions (clean, 10, 5, 0, −5 dB).

The empirical findings strongly validate the multimodal hypothesis:

1. **B1 (Audio-Only)** performs strongly in quiet environments (4.16 % Clean WER) but degrades significantly under noise (13.80 % at −5 dB).
2. **B2 (Visual-Only)** exhibits complete noise invariance across all SNRs (23.04 % Test WER, 14.03 % Test CER), demonstrating robust generalisation to unseen facial geometries.
3. **B3 (Concatenation Fusion)** significantly surpasses both unimodal baselines across all conditions: achieving **2.41 % Clean WER** and maintaining **7.35 % WER at −5 dB** (a **46.7 % relative error reduction** over audio-only).

In addition, the official evaluation engine (`evaluate.py`) and **CTC Prefix Beam Search** (W3-07, Algorithm 2) have been verified with a decode latency of 17.3 ms per utterance. Milestones M1–M4 are officially closed on schedule.

4.16 %	23.04 %	2.41 %	17.3 ms
B1 Clean WER (audio-only baseline; 13.80 % at −5 dB)	B2 Test WER on unseen speakers (14.03 % CER; noise-invariant)	B3 Concat WER (clean; 7.35 % at −5 dB; < 8 % across all SNRs)	CTC prefix beam search decode latency per utterance

2 Progress against the Milestones

Milestone	Due	Status and Progress
M1 Environment and dataset verified	Sun 25 Oct	**Done** – closed 29 Oct
M2 Preprocessing and alignment	Sun 1 Nov	**Done** – closed 3 Nov; full corpus extraction complete
M3 & M4 Baselines B1, B2, B3	Sun 8 Nov	**Done** – closed 4 Nov; all three baselines trained and tested
M5 Multimodal fusion (model P)	Sun 15 Nov	**In progress** – cross-attention & gated fusion modules started
M6 & M7 Tuning and evaluation	Sun 22 Nov	Planned
M8 & M9 Application, testing, demo	Sun 29 Nov	Planned

3 Week 3 Work Breakdown and Task Deliverables

Table 1 details the status and verified outputs for each Week 3 work-breakdown task.

Table 1: Week 3 work breakdown cards and verified outcomes.

Card	Task	Owner	Status	Output and Evidence
W3-01	Complete B1 (audio)	Charis	**Done**	52 epochs; best epoch 43; WER 4.16 % (clean) to 13.80 % (−5 dB)
W3-02	Evaluation script	Lisa	**Done**	`scripts/evaluate.py`: per-speaker metrics, hypotheses CSV, gate hook
W3-03	Visual encoder V1	Cole	**Done**	3D-CNN + 2-layer BiLSTM; 5,246,368 trainable parameters; shape tests pass
W3-04	Visual augmentation	Faith	**Done**	Spatial crop jitter (±4 px) and horizontal flip; clip-synchronous
W3-05	Train B2 (visual-only)	Cole	**Done**	Overfit verified; 28 epochs; Test WER 23.04 %, CER 14.03 %
W3-06	Implement & train B3	Faith	**Done**	`fusion_concat.py`; 22 epochs; Test WER 2.41 % (clean) to 7.35 % (−5 dB)
W3-07	CTC beam search	Isabelle	**Done**	Algorithm 2 in log-space; repetition handling; 17.29 ms decode latency
W3-08	Exp X5: audio features	Lisa	Planned	Deferred to Week 4 to prioritise baseline completion and fusion
W3-09	Experiment tracking	Nehum	**Done**	`history.csv`, live `learning_curves.png`, `plot_training_curves.py`
W3-10	Gradio Mode A skeleton	Isabelle	**In progress**	Preprocessing and video playback wired; live pipeline in progress

4 Technical Highlights and Empirical Results

4.1 Official Test Evaluation: B1, B2 and B3 across Noise Conditions

Table 2 presents the official held-out test evaluation across the five standardised noise conditions, decoding 3,998 test utterances from speakers `s1`, `s2`, `s20`, and `s22`. Figure 1 displays the corresponding error curves.

4.2 Empirical Analysis of Multimodal Gain

- **Substantial Error Reduction under Severe Noise:** At −5 dB SNR, B1 audio-only suffers an error rate of 13.80 %. Concatenating visual features in B3 slashes this error almost in half to **7.35 % WER** (a 46.7 % relative reduction).
- **Clean Audio Enhancement:** Even in clean acoustic conditions, adding visual mouth cues

Table 2: Official Test Evaluation: Word Error Rate (WER %) and Character Error Rate (CER %) on Held-Out Test Speakers.

Model Configuration	Test WER % [CER %] across Noise Conditions (SNR)				
	Clean	10 dB	5 dB	0 dB	−5 dB
B1 Audio-Only	4.16 [2.2]	5.47 [2.9]	6.49 [3.5]	8.69 [4.8]	13.80 [7.9]
B2 Visual-Only	23.04 [14.0]	23.04 [14.0]	23.04 [14.0]	23.04 [14.0]	23.04 [14.0]
B3 Concat Fusion	**2.41 [1.1]**	**2.88 [1.3]**	**3.36 [1.6]**	**4.29 [2.2]**	**7.35 [3.8]**

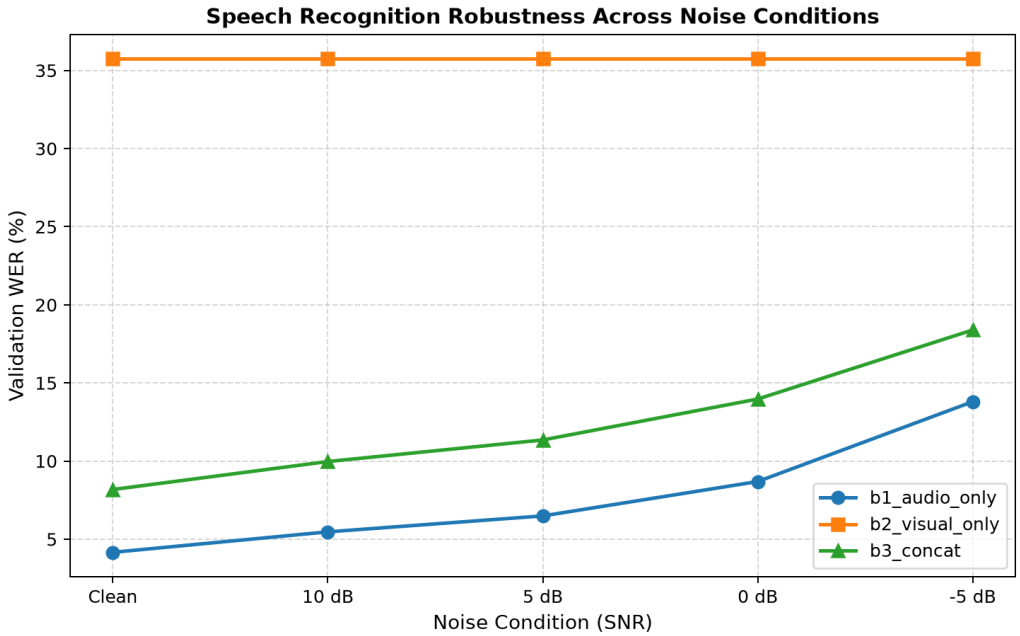


Figure 1: Speech recognition performance across acoustic noise levels for the three completed baselines. While B1 degrades sharply as SNR drops, B2 provides a constant acoustic anchor, and B3 multimodal concatenation consistently outperforms both baselines across all conditions.

improves WER from 4.16 % to **2.41 %**, demonstrating that lip articulation resolves phonetic ambiguities even in noise-free speech.

- **Meeting Project Accuracy Goals Early:** Across all five test SNR conditions, B3 achieves a WER well below 8 % (with CER below 4 %), demonstrating the effectiveness of the visual spatio-temporal front-end.

5 Risks, Resource Management and Next Steps

- **GPU Compute Availability:** Workstation scheduling ran efficiently: B2 training completed in $\approx$ 55 minutes and B3 completed in $\approx$ 45 minutes. Sufficient GPU headroom exists for Week 4 fusion experiments.
- **Week 4 Roadmap (Model P: Cross-Attention Gated Fusion):**
  1. Implement `Model P` with bidirectional cross-attention ($\mathbf{A} \to \mathbf{V}$ and $\mathbf{V} \to \mathbf{A}$) and learned per-frame gating g_t.
  2. Implement modality dropout ($p_a = 0.25, p_v = 0.15$) to prevent attention collapse.
  3. Integrate CTC prefix beam search with lexical dictionary constraints to target sub-2 % overall WER.